

The Finite-Type Diametral-Dimension Question Has a Known Negative Answer

Literature-status packet for arXiv:1908.01838

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Status: literature already answered. The later paper arXiv:2210.13593 explicitly answers the finite-type part of Question 1.1 of arXiv:1908.01838 by a counterexample.

1 Original question

In *Some Remarks on Diametral Dimension and Approximate Diametral Dimension of Certain Nuclear Fréchet Spaces* [1], Doğan asks in Question 1.1 (PDF page 2): for a nuclear Fréchet space E satisfying $\underline{DN}$ and Ω , if its diametral dimension $\Delta(E)$ coincides with that of a power series space, must its approximate diametral dimension $\delta(E)$ also coincide, and conversely?

The source proves a full affirmative equivalence for infinite-type power series spaces. For finite type it proves

$$\delta(E) = \delta(\Lambda_1(\varepsilon)) \iff [\Delta(E) = \Delta(\Lambda_1(\varepsilon)) \text{ and } E \text{ has a prominent bounded set}]$$

(Theorem 4.8, PDF page 11). Thus the remaining issue is whether equality of the finite-type diametral dimensions alone forces the equality of approximate diametral dimensions, or equivalently forces a prominent bounded set.

2 Explicit later answer

The later paper *On Power Series Subspaces of Certain Nuclear Fréchet Spaces* [2] repeats the earlier problem as Question 1.2 and states on PDF page 2 that the finite-type answer is negative. For every unstable finitely nuclear exponent sequence α , it constructs a nuclear Fréchet Köthe space K_α satisfying $\underline{DN}$ and Ω .

Propositions 3.6 and 3.7 (PDF pages 20–22) prove, with $\varepsilon_n = \alpha_{n+1}$,

$$\Delta(K_\alpha) = \Delta(\Lambda_1(\varepsilon)), \quad \delta(K_\alpha) \neq \delta(\Lambda_1(\varepsilon)).$$

Remark 3.8 explicitly identifies these two propositions as a negative answer to Question 1.2. Theorem 4.8 (PDF page 26) records the equivalent obstruction: K_α has no prominent bounded set. The answering paper cites the original paper and clearly identifies the question it resolves.

3 Classification and scope

This is a separate later paper explicitly answering the source question, so the correct run classification is `literature_already_answered`, not a new counterexample. The original infinite-type equivalence and the finite-type reverse implication remain intact. The counterexample refutes only the unrestricted finite-type implication

$$\Delta(E) = \Delta(\Lambda_1(\varepsilon)) \implies \delta(E) = \delta(\Lambda_1(\varepsilon)).$$

References

- [1] N. Doğan, *Some Remarks on Diametral Dimension and Approximate Diametral Dimension of Certain Nuclear Fréchet Spaces*, Bull. Belg. Math. Soc. Simon Stevin **27** (2020), 353–368; arXiv:1908.01838v2.
- [2] N. Doğan, *On Power Series Subspaces of Certain Nuclear Fréchet Spaces*, arXiv:2210.13593v1 (2022).